

JOAN GRAU NOËL

Site Reliability Engineer | Distributed Systems & Production Reliability

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PROFILE

Production-focused Site Reliability Engineer at Google working on globally distributed serverless infrastructure. Leads incident response, resilience, observability, capacity, and efficiency initiatives across Cloud Run, App Engine, and Cloud Functions. Combines Go and Python automation with C++ debugging and network troubleshooting to improve availability, accelerate root-cause analysis, and reduce operational toil.

EXPERIENCE

Google — Site Reliability Engineer / Software Engineer

April 2022–Present | London, United Kingdom

Site Reliability Engineer for Google's globally distributed Serverless Platform, serving O(millions) QPS across Cloud Run, App Engine, Cloud Functions, and internal workloads in many regions. Work across many binaries and development teams, covering traffic routing, ingress and egress networking, load balancing, low-latency multi-tenant serving, compute serving and scaling, sandboxed containers, container file serving, control-plane APIs, and databases.

- **Incident leadership:** Led response for approximately 20 major customer-facing outages and the majority of approximately 100 incidents overall, coordinating mitigation and follow-up across engineering teams.
- **Observability:** Led distributed tracing adoption across five software engineering teams and 10+ binaries, connecting request paths across asynchronous Spanner and Pub/Sub boundaries; reduced p90 time to debug and resolve incidents investigated using tracing by approximately 60%.
- **Automation:** Built a customer-facing quota broker using real-time capacity signals, combining automatic approvals with decision support for manual reviews; reduced associated operational toil for SRE and development teams by approximately 20–30%.
- **Capacity engineering:** Architected and deployed a supply-demand forecasting platform adopted by four partner teams, combining stockout prediction, forecast-skew detection, and capacity modelling.
- **Fleet efficiency:** Designed a regional scheduling migration to consolidate duplicate capacity buffers and improve fleet efficiency.
- **Resilience:** Built a continuous indicator of regional N+1 capacity coverage and led remediation of regions below target. Developed a drain simulator to evaluate failover capacity across zone, metro, and disaster-domain failures.
- **Technical leadership:** Own the team's capacity analytics service and set technical direction with tech leads and contributing engineers. Conduct production readiness reviews for autoscaling and deployment features, and mentor junior SREs and software engineers.

EARLIER EXPERIENCE

Google — Technical Solutions Engineer

March 2021–August 2022 | Dublin, Ireland

- Investigated enterprise serverless escalations involving architecture, networking, and performance; partnered with software and reliability engineers to identify platform issues, recurring failure modes, and long-term remediation plans.
- Built diagnostic scripts, automation tools, and troubleshooting playbooks to streamline issue reproduction and investigation. Translated customer incidents into observability improvements and product engineering priorities.

Amazon Web Services — Cloud Support Engineer, DevOps

July 2020–March 2021 | Dublin, Ireland

- Diagnosed production issues through reproduction, root-cause analysis, and systematic debugging; partnered with service engineering teams on complex customer escalations.
- Advised enterprise customers on scalability, reliability, and operational practices across Amazon EKS, Amazon ECS, Lambda, CloudWatch, and AWS CI/CD services.

SYSTEMS ENGINEERING & DEBUGGING

- **C++ diagnostics:** Investigated segmentation faults, memory leaks, race conditions, and performance regressions using heap and core dumps. Added instrumentation to C++ binaries and used distributed tracing to identify cross-service latency regressions.
- **Networking:** Investigated eBPF and NAT gateway incidents using tcpdump packet captures and low-level network debugging. Performed NAT capacity management and planning.
- **Production engineering:** Incident command, postmortems, automated capacity decisions, production safeguards, production readiness reviews, and technical mentoring.

TECHNICAL SKILLS

Languages: Python, Go, C++ (primary); Java, Kotlin (additional); SQL; Bash/Shell.

Cloud & platform: Google Cloud (Cloud Run, App Engine, Cloud Functions, Spanner, Pub/Sub); AWS (EKS, ECS, Lambda, CloudWatch, AWS CI/CD); Kubernetes; Docker; Protocol Buffers.

Systems & networking: Distributed systems, low-latency multi-tenant serving, traffic routing, load balancing, ingress and egress networking, NAT gateways, eBPF.

Reliability & analytics: Distributed tracing, SLIs/SLOs, monitoring and alerting, capacity forecasting, N+1 redundancy, failover modelling, SQL data pipelines, forecasting, and regression analysis.

EDUCATION & PROJECT

Universitat Pompeu Fabra — Audiovisual Systems Engineering

2014–2018 | Barcelona, Spain

Thesis — Audio source separation: Adapted a CNN-based PyTorch architecture for monaural audio source separation and applied data augmentation to improve model output. [Source code and thesis](#)